SESSION 1 REFERENCE GUIDE: PYTHON BASICS & COMPUTATIONAL LOGIC

This guide provides a reference for the core Python syntax, data constructs, and analytical workflows introduced in Session 1.

1. Core Python Syntax & Primitives

Variables

A variable stores data in computer memory using an assignment operator (=). Variable names should be descriptive and use lower_snake_case.

```python
# standard variable assignments
department_name = "Analytics & Insights" # String (str)
staff_count = 5 # Integer (int)
benchmark_salary = 75000.0 # Float (float)
is_budget_approved = False # Boolean (bool)
```

Basic Arithmetic Operators

Operator	Operation	Example Code	Result
+	Addition	72000.0 + 81000.0	153000.0
-	Subtraction	81000.0 - 75000.0	6000.0
*	Multiplication	5 * 75000.0	375000.0
/	Division	385000.0 / 5	77000.0

2. Python Lists & Indexing

A list is an ordered collection of values enclosed in square brackets [].

```python
# Creating a list of float numbers
salaries = [72000.0, 81000.0, 75000.0, 69000.0, 88000.0]
```

Indexing Rules

Python uses **zero-based indexing**, meaning the first element is located at index 0.

```python
first_salary  = salaries[0]  # Accesses 72000.0
second_salary = salaries[1]  # Accesses 81000.0
last_salary   = salaries[4]  # Accesses 88000.0
```

3. Built-In Analytical Functions

Python provides built-in summary functions that operate on numerical lists:

```python
numbers = [10.0, 20.0, 30.0, 40.0]

# sum(): Adds all numerical elements in a list
total_sum = sum(numbers)  # Result: 100.0

# len(): Returns the count of elements in a list
element_count = len(numbers)  # Result: 4

# Computing Mean: Divide sum by length
mean_value = sum(numbers) / len(numbers)  # Result: 25.0
```

4. The Junior Analyst Workflow Framework

When assigned an analytical task, follow this 4-step framework:

```
[1. Clarify Request] --> [2. Audit & Structure] --> [3. Compute Metrics] -->
[4. Interpret & Recommend]
```

1. **Clarify Request:** Identify target metrics and benchmark values.
2. **Audit & Structure:** Store data in appropriate data types (floats, lists).
3. **Compute Metrics:** Calculate sum, count, mean, and variances.
4. **Interpret & Recommend:** Translate output into business implications (e.g., percentage over baseline).